Name: Prashanth Raghavendra Rao

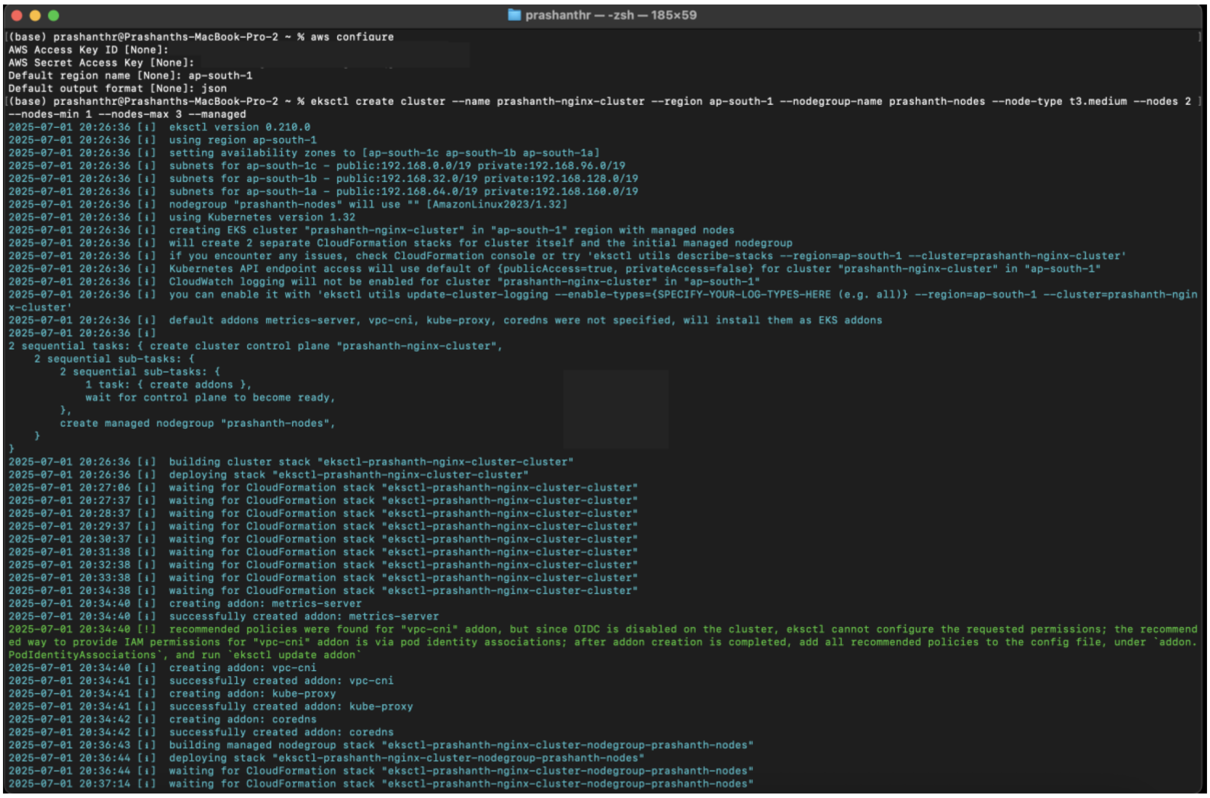
Task: K8s EKS 2

Date: 01/07/2025

Task Description:

Create the K8s EKS,further you have to do the deployment of the Nginx application and access the application outside the cluster.

1. Configure AWS CLI and Create Cluster



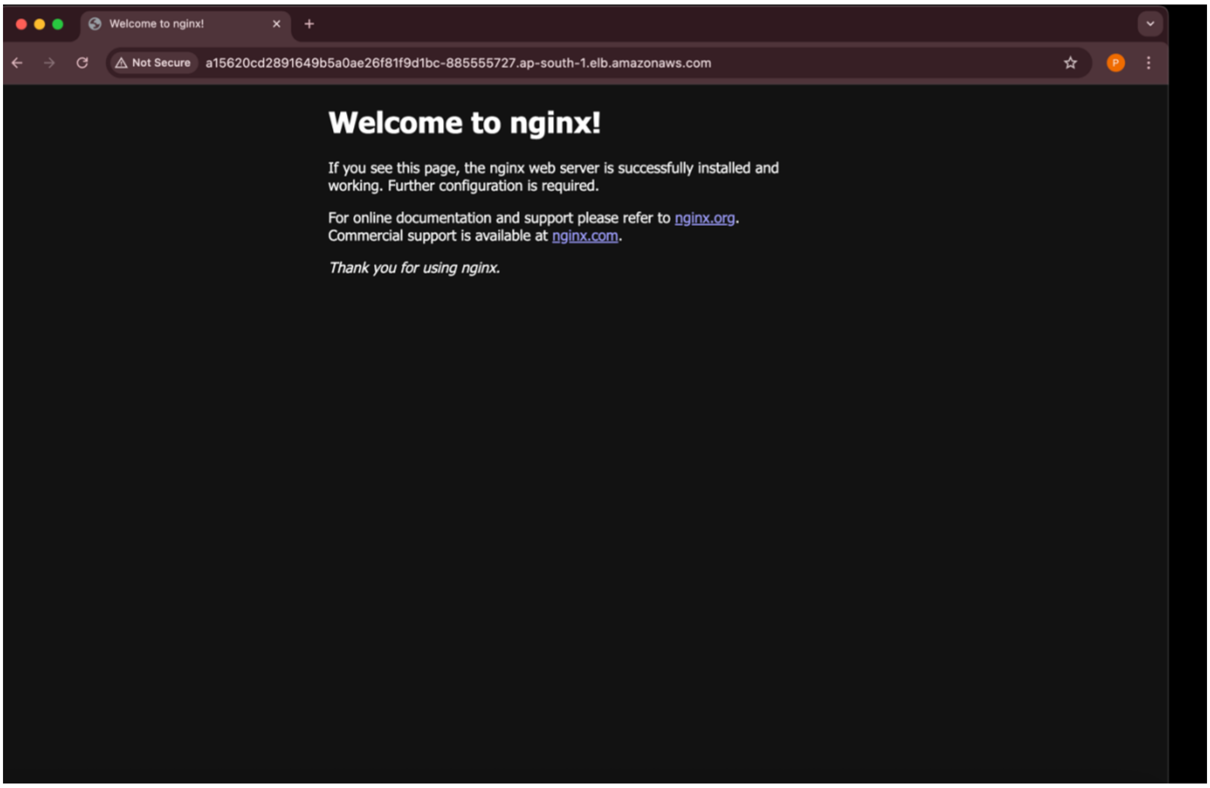
```
prashanthr — -zsh — 185×59
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % aws configure
AWS Access Key ID [None]: [redacted]
AWS Secret Access Key [None]: [redacted]
Default region name [None]: ap-south-1
Default output format [None]: json
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % eksctl create cluster --name prashanth-nginx-cluster --region ap-south-1 --nodegroup-name prashanth-nodes --node-type t3.medium --nodes 2
--nodes-min 1 --nodes-max 3 --managed
2025-07-01 20:26:36 [ℹ]  eksctl version 0.210.0
2025-07-01 20:26:36 [ℹ]  using region ap-south-1
2025-07-01 20:26:36 [ℹ]  setting availability zones to [ap-south-1c ap-south-1b ap-south-1a]
2025-07-01 20:26:36 [ℹ]  subnets for ap-south-1c - public:192.168.0.0/19 private:192.168.96.0/19
2025-07-01 20:26:36 [ℹ]  subnets for ap-south-1b - public:192.168.32.0/19 private:192.168.128.0/19
2025-07-01 20:26:36 [ℹ]  subnets for ap-south-1a - public:192.168.64.0/19 private:192.168.160.0/19
2025-07-01 20:26:36 [ℹ]  nodegroup "prashanth-nodes" will use "" [AmazonLinux2023/1.32]
2025-07-01 20:26:36 [ℹ]  using Kubernetes version 1.32
2025-07-01 20:26:36 [ℹ]  creating EKS cluster "prashanth-nginx-cluster" in "ap-south-1" region with managed nodes
2025-07-01 20:26:36 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2025-07-01 20:26:36 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=prashanth-nginx-cluster'
2025-07-01 20:26:36 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "prashanth-nginx-cluster" in "ap-south-1"
2025-07-01 20:26:36 [ℹ]  CloudWatch logging will not be enabled for cluster "prashanth-nginx-cluster" in "ap-south-1"
2025-07-01 20:26:36 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=ap-south-1 --cluster=prashanth-ngin
x-cluster'
2025-07-01 20:26:36 [ℹ]  default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2025-07-01 20:26:36 [ℹ]
2 sequential tasks: { create cluster control plane "prashanth-nginx-cluster",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "prashanth-nodes",
    }
}
2025-07-01 20:26:36 [ℹ]  building cluster stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:26:36 [ℹ]  deploying stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:27:06 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:27:37 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:28:37 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:29:37 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:30:37 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:31:38 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:32:38 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:33:38 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:34:38 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-cluster"
2025-07-01 20:34:40 [ℹ]  creating addon: metrics-server
2025-07-01 20:34:40 [ℹ]  successfully created addon: metrics-server
2025-07-01 20:34:40 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommend
ed way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under `addon.
PodIdentityAssociations`, and run `eksctl update addon`
2025-07-01 20:34:40 [ℹ]  creating addon: vpc-cni
2025-07-01 20:34:41 [ℹ]  successfully created addon: vpc-cni
2025-07-01 20:34:41 [ℹ]  creating addon: kube-proxy
2025-07-01 20:34:41 [ℹ]  successfully created addon: kube-proxy
2025-07-01 20:34:42 [ℹ]  creating addon: coredns
2025-07-01 20:34:42 [ℹ]  successfully created addon: coredns
2025-07-01 20:36:43 [ℹ]  building managed nodegroup stack "eksctl-prashanth-nginx-cluster-nodegroup-prashanth-nodes"
2025-07-01 20:36:44 [ℹ]  deploying stack "eksctl-prashanth-nginx-cluster-nodegroup-prashanth-nodes"
2025-07-01 20:36:44 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-nodegroup-prashanth-nodes"
2025-07-01 20:37:14 [ℹ]  waiting for CloudFormation stack "eksctl-prashanth-nginx-cluster-nodegroup-prashanth-nodes"
```

2. Deployment and Service

```
prashanthr — -zsh — 185×59
2025-07-01 20:40:18 [ℹ]  node "ip-192-168-59-154.ap-south-1.compute.internal" is ready
2025-07-01 20:40:18 [✔]  created 1 managed nodegroup(s) in cluster "prashanth-nginx-cluster"
2025-07-01 20:40:19 [ℹ]  kubectl command should work with "/Users/prashanthr/.kube/config", try 'kubectl get nodes'
2025-07-01 20:40:19 [✔]  EKS cluster "prashanth-nginx-cluster" in "ap-south-1" region is ready
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % eksctl get cluster --region ap-south-1
NAME                    REGION          EKSCTL CREATED
prashanth-nginx-cluster ap-south-1      True
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % aws eks update-kubeconfig --region ap-south-1 --name prashanth-nginx-cluster
Added new context arn:aws:eks:ap-south-1:235391889675:cluster/prashanth-nginx-cluster to /Users/prashanthr/.kube/config
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get nodes
NAME                                            STATUS   ROLES    AGE     VERSION
ip-192-168-19-196.ap-south-1.compute.internal   Ready    <none>   2m52s   v1.32.3-eks-473151a
ip-192-168-59-154.ap-south-1.compute.internal   Ready    <none>   2m54s   v1.32.3-eks-473151a
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % nano prashanth-nginx-deployment.yaml
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % cat prashanth-nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl apply -f prashanth-nginx-deployment.yaml
deployment.apps/nginx-deployment created
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get pods
NAME                               READY   STATUS              RESTARTS   AGE
nginx-deployment-96b9d695-hrwm8   0/1     ContainerCreating   0          7s
nginx-deployment-96b9d695-wgw8k   0/1     ContainerCreating   0          7s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % nano prashanth-nginx-service.yaml
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % cat prashanth-nginx-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
  type: LoadBalancer

(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
nginx-deployment-96b9d695-hrwm8   1/1     Running   0          42s
nginx-deployment-96b9d695-wgw8k   1/1     Running   0          42s
```

3. Deploy and Expose

```
prashanthr — -zsh — 185×59
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl apply -f prashanth-nginx-deployment.yaml
deployment.apps/nginx-deployment created
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get pods
NAME                               READY   STATUS              RESTARTS   AGE
nginx-deployment-96b9d695-hrwm8   0/1     ContainerCreating   0          7s
nginx-deployment-96b9d695-wgw8k   0/1     ContainerCreating   0          7s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % nano prashanth-nginx-service.yaml
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % cat prashanth-nginx-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
  type: LoadBalancer

(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
nginx-deployment-96b9d695-hrwm8   1/1     Running   0          42s
nginx-deployment-96b9d695-wgw8k   1/1     Running   0          42s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl apply -f prashanth-nginx-service.yaml
service/nginx-service created
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get svc
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                     443/TCP        11m
nginx-service   LoadBalancer   10.100.128.242   a15620cd2891649b5a0ae26f81f9d1bc-885555727.ap-south-1.elb.amazonaws.com   80:30511/TCP   6s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get svc
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                     443/TCP        13m
nginx-service   LoadBalancer   10.100.128.242   a15620cd2891649b5a0ae26f81f9d1bc-885555727.ap-south-1.elb.amazonaws.com   80:30511/TCP   104s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get svc
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                     443/TCP        13m
nginx-service   LoadBalancer   10.100.128.242   a15620cd2891649b5a0ae26f81f9d1bc-885555727.ap-south-1.elb.amazonaws.com   80:30511/TCP   2m30s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl expose deployment nginx --type=LoadBalancer --port=80
Error from server (NotFound): deployments.apps "nginx" not found
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl expose deployment prashanth-nginx --type=LoadBalancer --port=80
Error from server (NotFound): deployments.apps "prashanth-nginx" not found
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl expose deployment prashanth-nginx-deployment --type=LoadBalancer --port=80
Error from server (NotFound): deployments.apps "prashanth-nginx-deployment" not found
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl expose deployment prashanth-nginx-service --type=LoadBalancer --port=80
Error from server (NotFound): deployments.apps "prashanth-nginx-service" not found
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl expose deployment nginx-service --type=LoadBalancer --port=80
Error from server (NotFound): deployments.apps "nginx-service" not found
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ % kubectl get svc
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                     443/TCP        15m
nginx-service   LoadBalancer   10.100.128.242   a15620cd2891649b5a0ae26f81f9d1bc-885555727.ap-south-1.elb.amazonaws.com   80:30511/TCP   4m7s
(base) prashanthr@Prashanths-MacBook-Pro-2 ~ %
```



4. Ensure Deletion